

HR Employee Attrition Analysis using Python

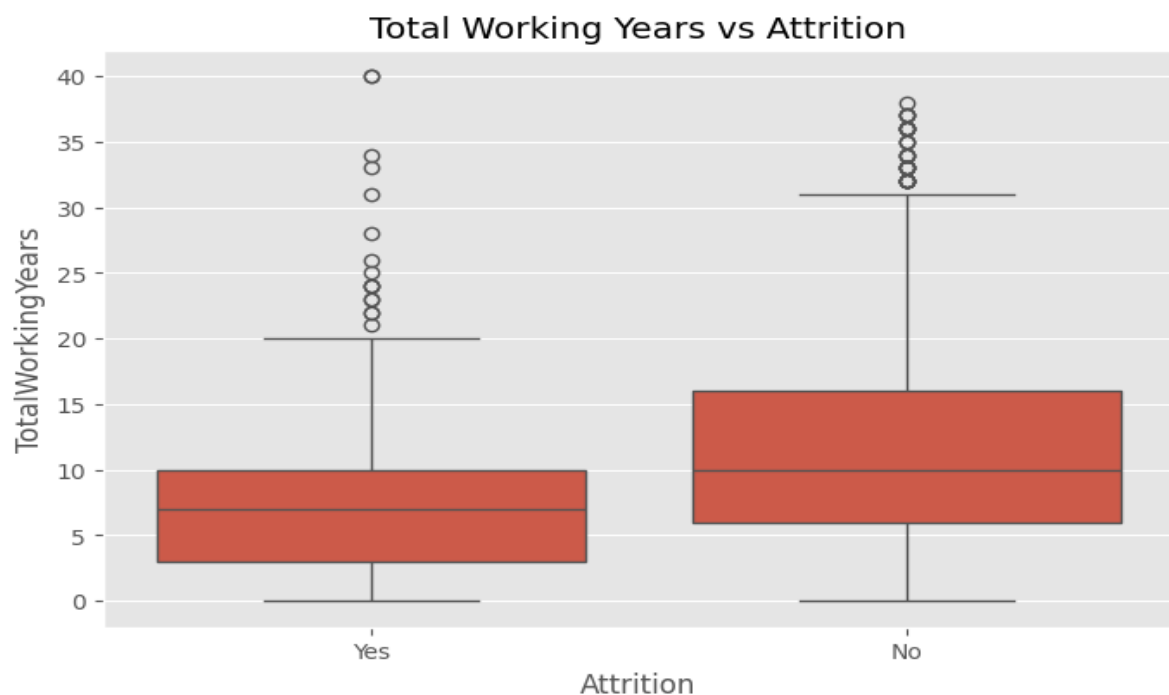
CodeAlpha Data Analytics Internship

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Introduction

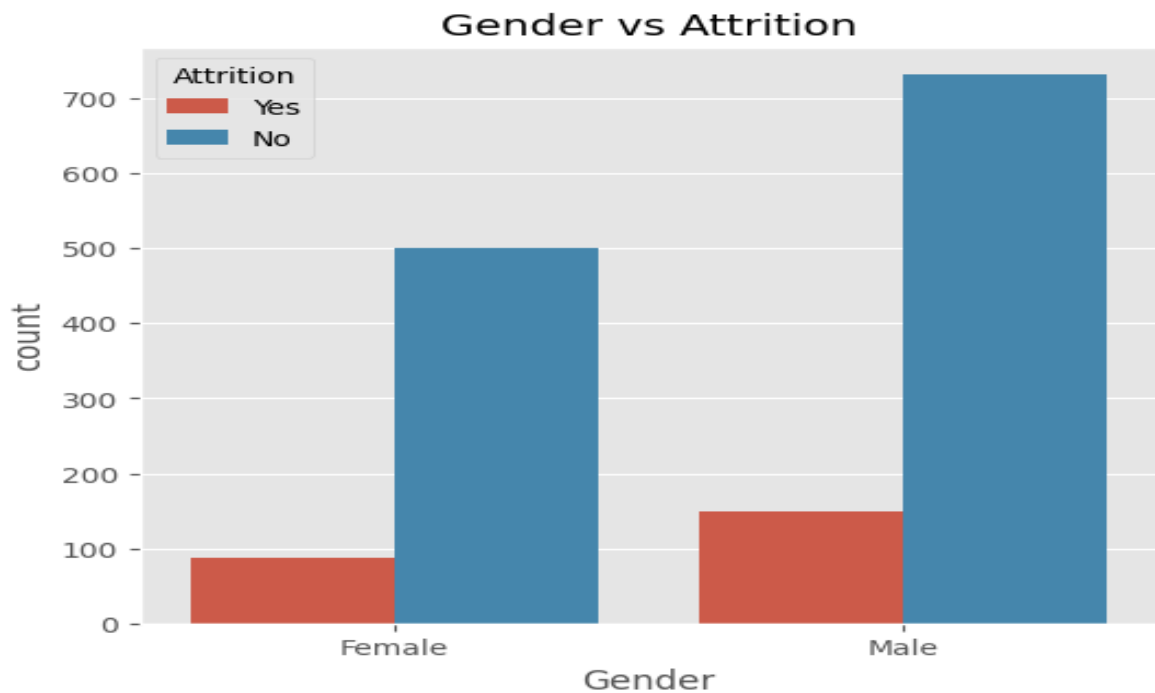
This report presents an exploratory data analysis of the IBM HR Employee Attrition dataset using Python, Pandas, Matplotlib and Seaborn. The objective is to identify key factors influencing employee attrition and provide business recommendations.

Department-wise Employee Attrition



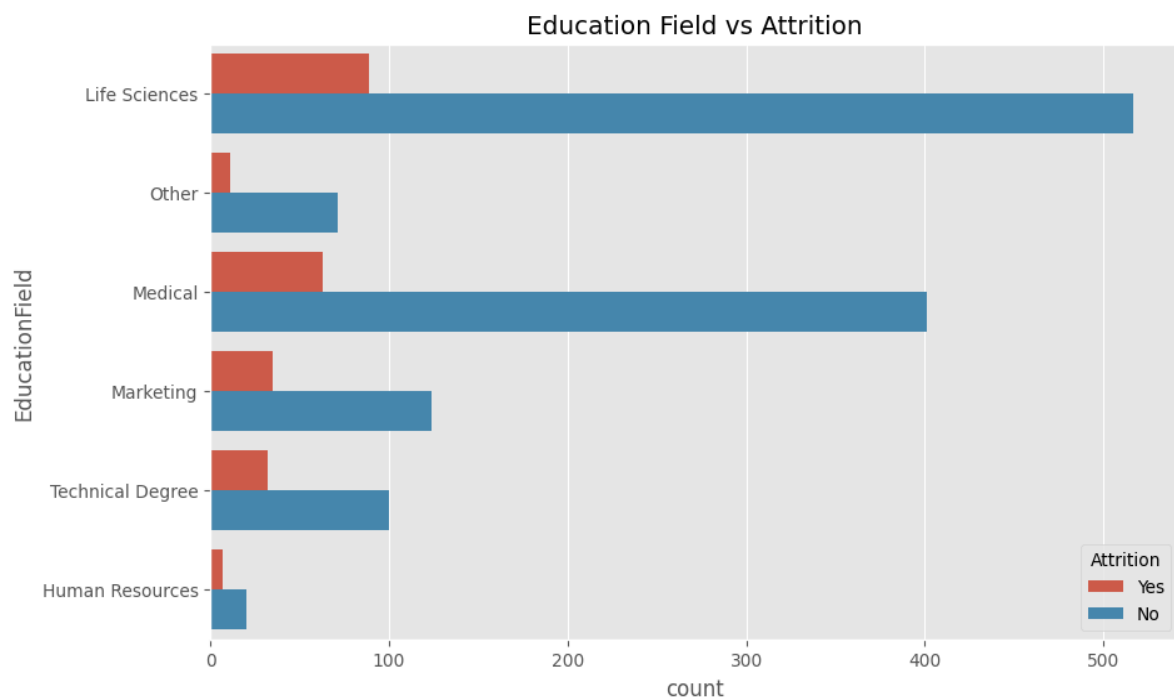
Insight: Research & Development has the highest number of employee exits because it has the largest workforce.

Department-wise Attrition Rate



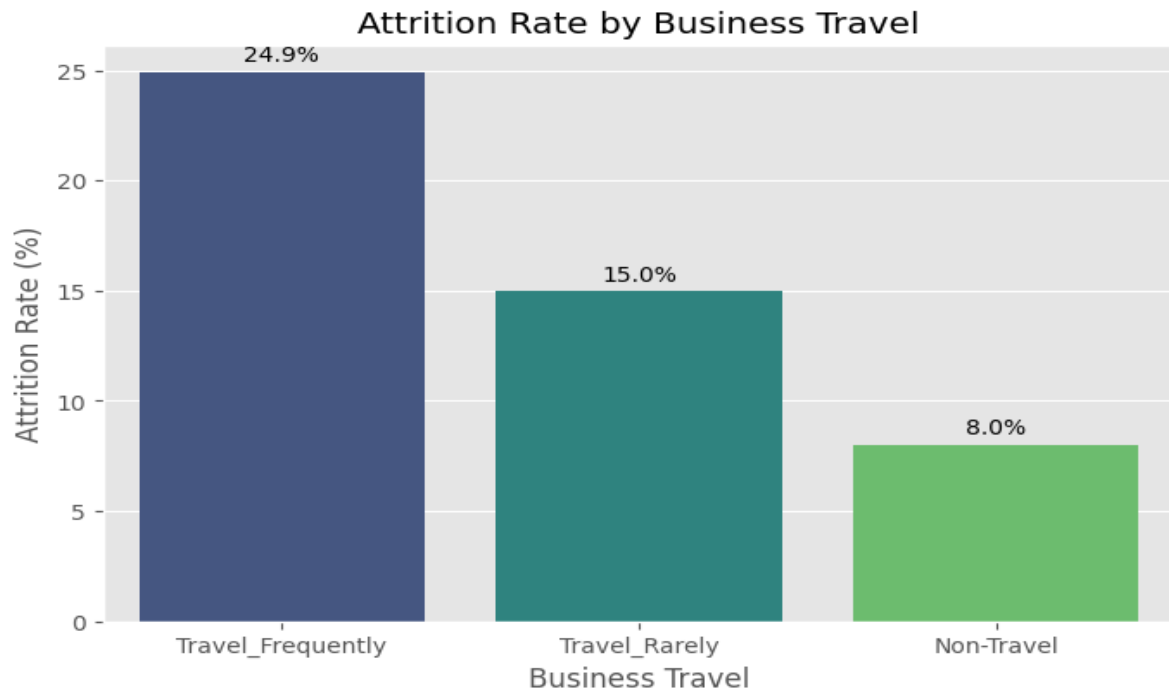
Insight: Sales has the highest attrition rate (20.6%), followed by Human Resources (19.0%) and R&D; (13.8%).

Age Distribution



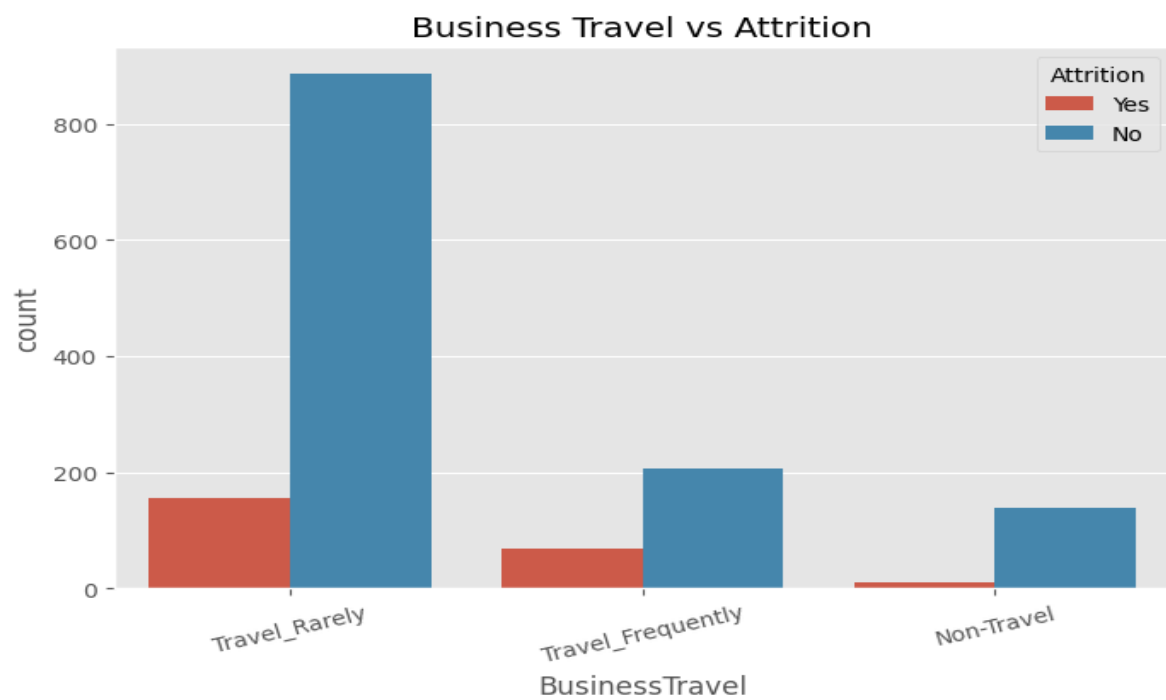
Insight: Employees aged 26–35 show the highest attrition counts.

Overtime vs Attrition



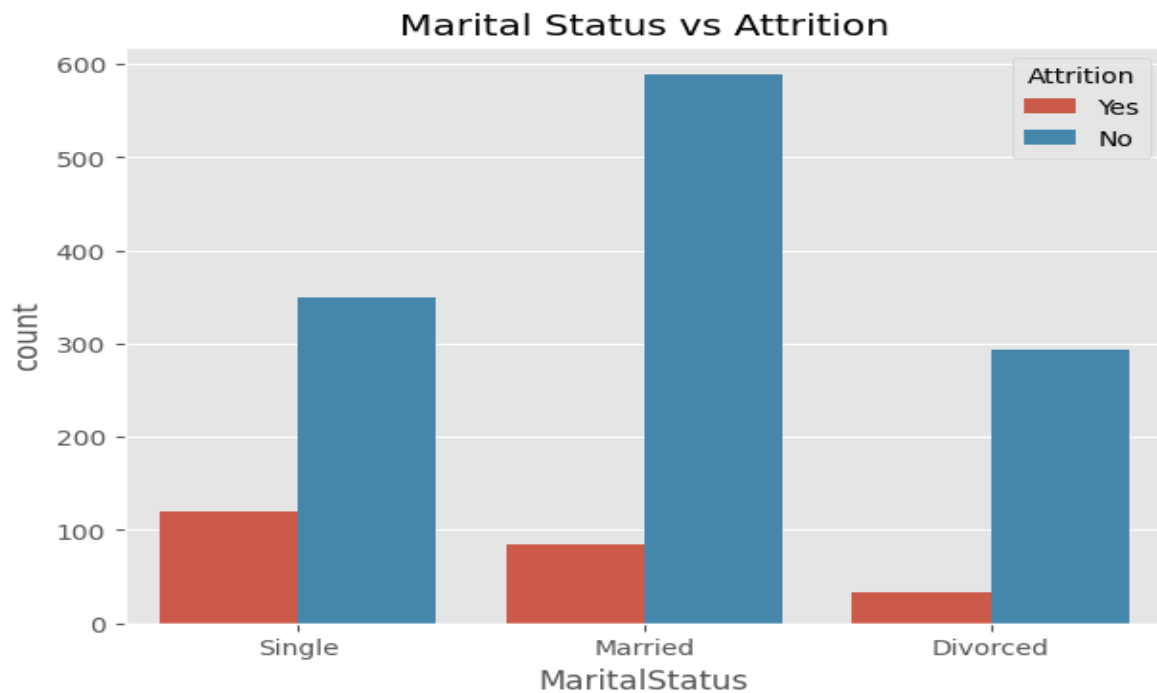
Insight: Employees working overtime are more likely to leave the organization.

Job Role



Insight: Laboratory Technicians and Sales Executives show relatively high turnover.

Monthly Income



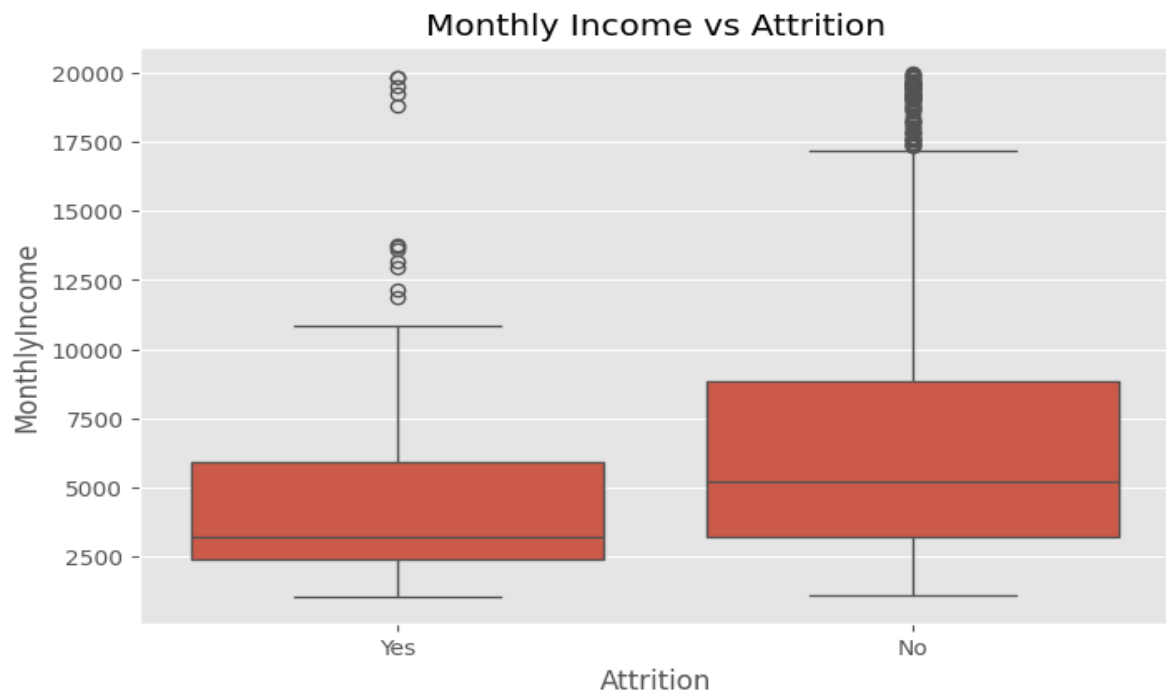
Insight: Employees leaving the company generally have lower monthly income.

Job Satisfaction



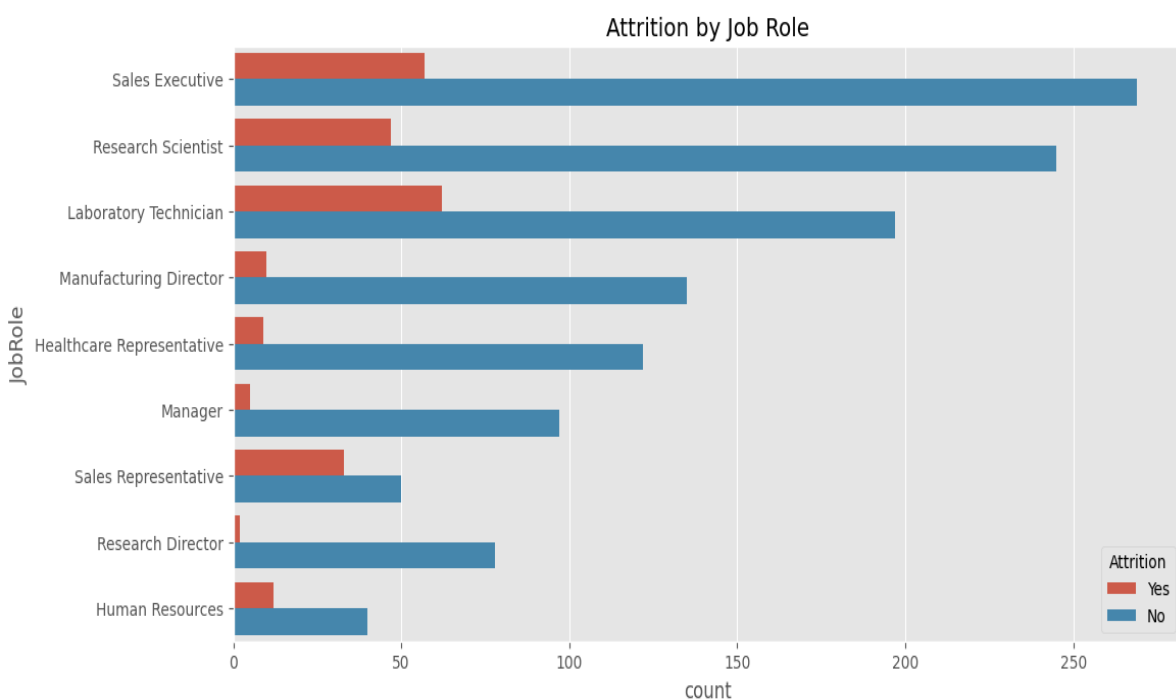
Insight: Lower job satisfaction is associated with higher attrition.

Marital Status



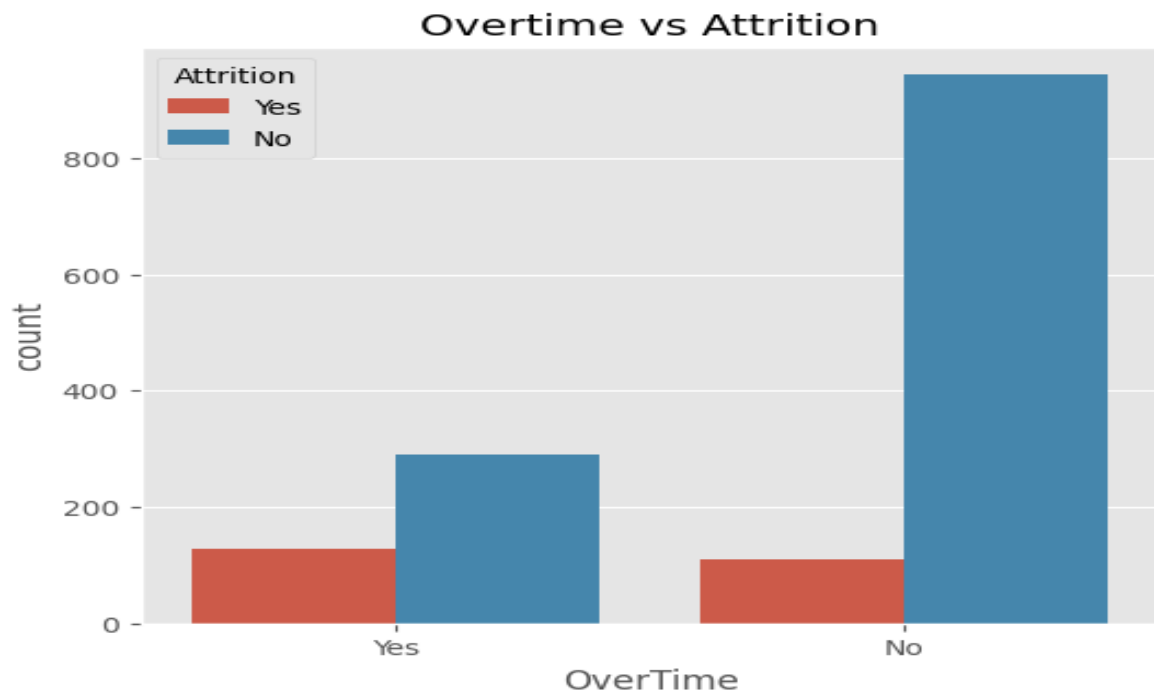
Insight: Single employees exhibit higher attrition than married and divorced employees.

Business Travel Count



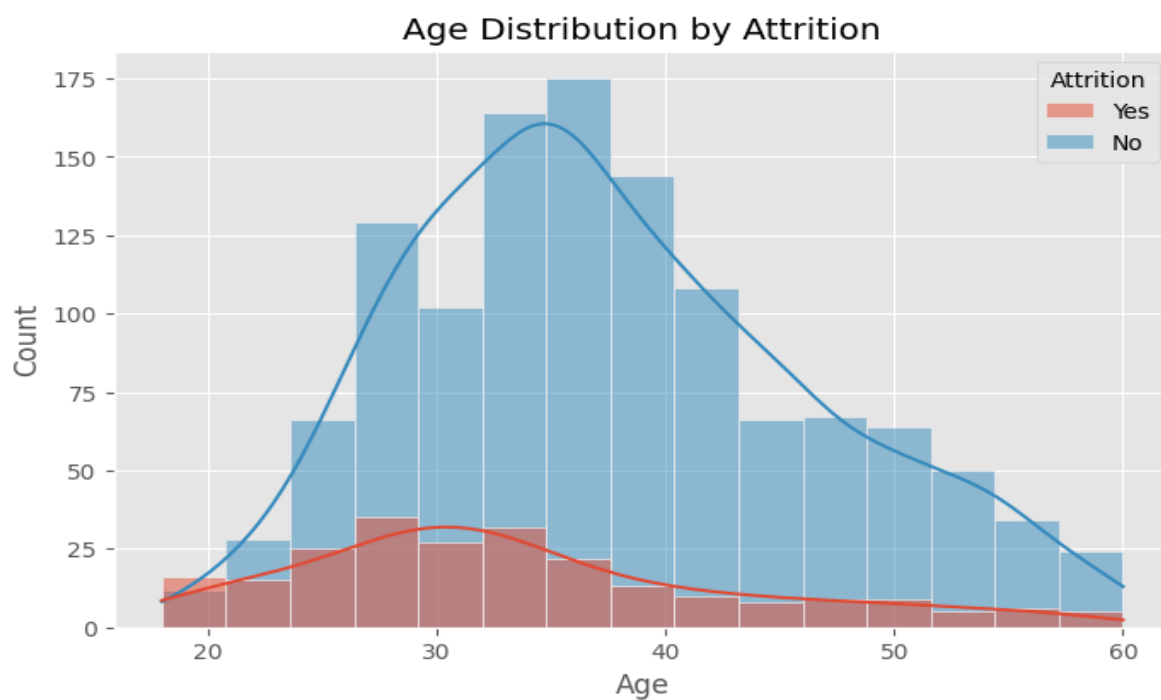
Insight: Travel_Rarely has the highest number of exits because it has the largest employee population.

Business Travel Rate



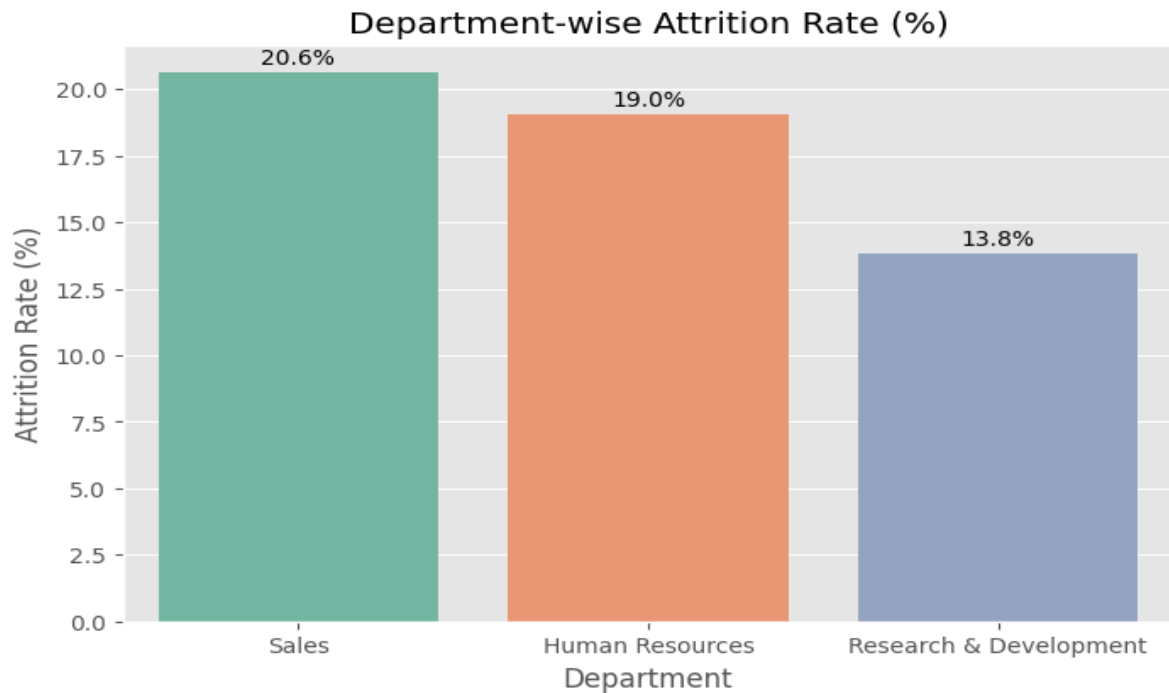
Insight: Travel_Frequently has the highest attrition rate (24.9%).

Education Field



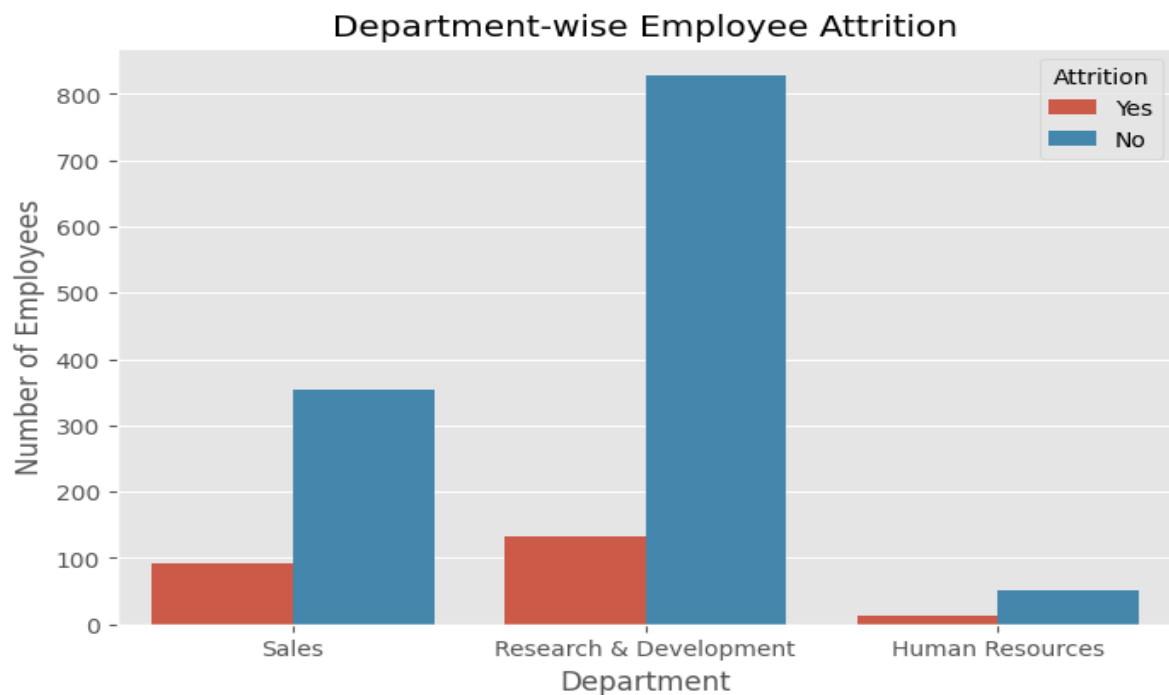
Insight: Life Sciences contributes the largest number of attrition cases.

Gender



Insight: Male employees have more attrition cases in absolute numbers.

Working Years



Insight: Employees with fewer total working years are more likely to leave.

Business Recommendations

- Prioritize retention initiatives in Sales due to its highest attrition rate.
- Investigate workload and engagement in the R&D; department.

- Reduce excessive overtime and improve work-life balance.
- Strengthen employee engagement and job satisfaction programs.
- Review compensation and career development opportunities for early-career employees.
- Use HR dashboards for continuous monitoring of attrition.

Conclusion

Employee attrition is influenced by multiple factors including department, overtime, business travel, compensation, job satisfaction and experience. Applying targeted HR strategies can improve retention and organizational performance.